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RESEARCH NOTE

Oscillatory EEG correlates of arithmetic strategy use in addition and subtraction

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Abstract Adults use different strategies in mental arithmetic. For example, they directly retrieve the answer from memory or calculate by means of procedural strategies. Despite growing insight into the hemodynamic and electrophysiological correlates of these strategies, the functional changes in the oscillatory brain dynamics during the use of these strategies remain unknown. In the present high-resolution electroencephalography (EEG) study, we analysed event-related synchronisation (ERS) and desynchronisation (ERD) in the theta and alpha bands while participants solved addition and subtraction problems, which displayed a high probability of retrieval or procedural strategy use. Findings revealed that arithmetic fact retrieval is reflected in left-hemispheric ERS in the theta band, whereas the application of procedural strategies is accompanied by bilateral parietooccipital ERD in the alpha band. The topographical and frequency specificity of the strategy effects provides a start for the development of electrophysiological indices of strategy use in arithmetic.

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Introduction

It is well established that educated adults use different strategies in mental arithmetic (e.g., Campbell and Xue 2001; LeFevre et al. 1996): they directly retrieve the answer from long-term memory or they calculate the answer by means of procedural strategies (e.g., decomposition of a problem into smaller facts). While retrieval is fast, requires minimal cognitive load and involves little or no magnitude processing, procedural strategies are more time-consuming and rely stronger on magnitude processing and working memory resources. Neuroimaging studies have recently started to unravel the brain correlates of these solution strategies. Functional magnetic resonance imaging (fMRI) studies have revealed that small single-digit arithmetic problems, which are likely to be solved by fact retrieval (see, e.g., Campbell and Xue 2001), elicited stronger activation in left-hemispheric language regions, including the supramarginal and angular gyri. In contrast, large problems, which are assumed to rely more on procedural strategies, were associated with stronger bilateral activation in frontal and parietal cortices, the latter being associated with magnitude processing (e.g., Dehaene et al. 2003; Grabner et al. 2007a; Stanescu-Cosson et al. 2000). Similarly, Grabner et al. (2009) recently showed that the application of retrieval strategies elicited stronger left angular gyrus activation, while procedural strategies evoked bilateral widespread fronto-parietal activation. The involvement of left-hemispheric language-related brain regions in calculation has been interpreted to reflect the demand on verbal coding of numbers or on the retrieval of verbally stored arithmetic

knowledge from memory (Dehaene et al. 2003). In addition to fMRI, event-related potential (ERP) studies have shed light on the time-course of brain activity during the application of different arithmetic strategies, pointing to an amplitude modulation of the late positive slow wave during procedural calculation but not during retrieval (Jost et al. 2004; Nunez-Pena et al. 2006).

Despite growing insight into the hemodynamic and electrophysiological correlates of arithmetic strategies, little is known about how the involved neuronal networks are formed and interact with each other during arithmetic. Therefore, we analysed induced EEG activity, which is a well-established way to investigate the dynamics of functional network formation during cognitive information processing (Bastiaansen and Hagoort 2003; Klimesch 1999; Pfurtscheller and Lopes da Silva 2005). In contrast to ERPs which reflect a time- and phase-locked response of the brain to a single event (time-locked) induced activity refers to functional changes in the oscillatory brain activity elicited by an event or task. Basically, the pattern of synchronisation and desynchronisation (ERD) of neuronal activity in different frequency bands is related to the coupling and uncoupling of functional networks in the brain (Bastiaansen et al. 2006; Klimesch et al. 2005; Neuper and Pfurtscheller 2001). The quantification of event-related synchronisation (ERS) and desynchronisation (ERD) has proved to be a useful method to determine functional changes in the oscillatory brain dynamics during cognitive information processing (for an overview, cf. Neuper and Klimesch 2006). This method calculates the percentage amount of band power increases (ERS) or decreases (ERD) in any frequency band from a pre-stimulus reference interval to an activation interval (Pfurtscheller and Lopes da Silva 2005).

Studies applying the ERS/ERD methodology have accumulated a large basis of empirical evidence suggesting a differential functional significance of different frequency bands depending on the scalp topography. Global and posterior increases in theta activity (ERS) have repeatedly been associated with tasks demanding memory encoding and retrieval (Burgess and Gruzelier 2000; Klimesch et al. 1997) and with retrieval of lexical-semantic information from long-term memory in particular (Bastiaansen et al. 2006). Higher theta ERS was observed to be related to the retrieval of lexical information while reading (Bastiaansen et al. 2005) and translating words (Grabner et al. 2007b). The functional specificity of alpha oscillations (around 8–12 Hz) is less clear as alpha ERD has been reported in response to various cognitive task demands (Neubauer et al. 2006). As the amount of alpha ERD has turned out to generally increase with task difficulty, it has been interpreted as an index of general cortical activation or invested cognitive resources (Pfurtscheller and Lopes da Silva 1999).

Studies focusing on oscillatory EEG activity during arithmetic are scarce. Earle et al. (1996) found higher left-hemispheric theta band power when participants solved arithmetic problems involving symbolic and verbal operators compared to a task in which they had to supply the correct arithmetic sign into an equation. This finding was interpreted to reflect fact retrieval. Harmony et al. (1999) reported, compared to a physical baseline, larger task-related activation changes during a complex arithmetic task [e.g., (24 + 39)/9] in the theta frequency range (around 3–4 Hz) in left parietooccipital regions (band power increases) and in the alpha frequency range (around 12 Hz) in the left parietal regions (band power decreases). Theta effects were interpreted to reflect sustained attention, whereas the alpha effect was associated with long-term retrieval. Recently, Micheloyannis et al. (2005) observed frontal theta power increases and parietal alpha power decreases during multiplication relative to viewing numbers.

Although the aforementioned studies have provided important first insights into the oscillatory correlates of arithmetic, their significance with respect to strategy use remains limited. First, the selection of the arithmetic problems was not designed to differentiate between retrieval versus procedural strategies, making it unclear which strategies were likely applied in the administered conditions. Second, these studies did not always distinguish between different operations, which may involve different strategies. Finally, these studies analysed fixed time periods (activation intervals) for all individuals and all task conditions. This may not capture the individual and task-specific oscillatory changes related to cognitive strategies appropriately. For example, when a constant activation interval of several seconds after stimulus onset is used, the activation interval in faster individuals or easier task conditions may cover not only problem solving but also other task-irrelevant cognitive processes.

The present study is the first to explore the ERS/ERD correlates of retrieval and procedural strategies in arithmetic taking into account the aforementioned limitations. We recorded a 64-channel EEG and analysed the topographical ERS/ERD patterns in the alpha and theta frequency range using individually and trial-wise defined activation periods (cf. Grabner et al. 2004, 2006; Neubauer et al. 2004). To distinguish retrieval and procedural strategies, we presented additions and subtractions that have been shown to display a high probability of retrieval or procedural strategy use. Retrieval problems had a sum or minuend below or equal to 10, which showed high frequencies of retrieval use in educated adults (95% in addition and 86% in subtraction; Campbell and Xue 2001). Problems with a high probability of procedural strategies involved two-digit calculations, to which the answer is unlikely to be directly retrieved from

memory. We expected the topographical ERS/ERD patterns in the alpha and theta band to differ between retrieval and procedural problems. Based on findings from previous EEG studies linking theta power increases with memory retrieval, we hypothesised that the solution of retrieval problems would be associated with higher theta ERS than the solution of procedural problems. The strong association between alpha ERD and task difficulty suggests that procedural problems would elicit more alpha ERD than retrieval problems.

In addition to the ERS/ERD correlates of different strategies, we also investigated potential neurophysiological differences between addition and subtraction. This goal is motivated by neuroimaging and neuropsychological findings suggesting that different operations draw on different codes and require the recruitment of both shared and separate neural resources (Dehaene and Cohen 1997; Fehr et al. 2007; Kong et al. 2005). The triple-code model (Dehaene and Cohen 1997) predicts that single-digit addition relies on a verbal code while subtraction involves more on a visuospatial quantity code. This stronger reliance on quantity processing in subtraction may be reflected in the alpha frequency range.

Method

Participants

Participants were 25 right-handed healthy university students without known calculation difficulties. Four participants were excluded because their performance accuracy was below 50%; one participant was excluded due to technical problems during EEG acquisition. The remaining sample of 20 adults comprised 11 males and 9 females (mean age = 22.60, SD = 3.10). The study was approved by the local ethics committee (Swiss Federal Institute of Technology Zurich, Switzerland). All participants gave written informed consent.

Materials and apparatus

Participants solved additions and subtractions that differentially drew on retrieval and procedural strategies. No zero-, one-, or tie problems (e.g., 6 + 6) were included. Retrieval problems had a sum or minuend between 2 and 10 (e.g., 4 + 2). All procedural problems comprised two-digit numbers with a sum or minuend between 11 and 60. Two types of procedural complexity were presented: complex procedural problems, which required carrying/borrowing (e.g., 53–26), and simple procedural problems, without carrying/borrowing (e.g., 24 + 31). Each problem type involved 48 additions and 48 subtractions, yielding a total of 288 trials. The position of the largest addend and the size of the subtrahend were counterbalanced.

Each EEG trial started with the presentation of a fixation point for 3,000 ms, followed by the arithmetic problem. The participant was instructed to solve the problem as correctly and quickly as possible and to speak the solution into a voice-activated device (voice-key) that was connected to the computer. The experimenter recorded the participants answer to the problem. The next trial followed after an inter-trial interval of 2,000 ms.

EEG was acquired through the BioSemi ActiveTwo system (BioSemi, Amsterdam, The Netherlands) from 64 scalp electrodes placed according to the extended 10–20 systems. An electrooculogram (EOG) was recorded from four additional electrodes; two placed horizontally at the outer canthi of both eyes, and two placed vertically below and above the inner canthus of the left eye. EEG and EOG signals were sampled at 256 Hz and filtered between DC and 128 Hz.

Procedure

The EEG recording started with a 3 min rest EEG during which the participant was instructed to open the eyes, close them, or to deliberately blink, roll and move the eyes. This sequence was required for the automatic reduction of eye movement artefacts. Next, participants solved the arithmetic problems which were presented in four runs. Runs were separated by short breaks in order to avoid participants losing their concentration. Within each run of 72 additions or 72 subtractions, all problem types (retrieval, simple procedural, complex procedural) were presented intermixed in pseudo-randomised sequence. The order of the runs was counterbalanced across participants. The EEG session took about 2 h.

Data analysis

EEG data were band-pass filtered between 0.5 and 45 Hz to eliminate slow-frequency and power-line contamination artefacts. EOG artefacts were automatically reduced by employing a regression method (Schlogl et al. 2007) based on the resting EEG sequence. Afterwards, all EEG trials were visually inspected for artefacts. The spatial information of the artefact-free EEG data was enhanced by applying a 3D spline surface Laplacian estimation (Babiloni et al. 1998). This procedure has turned out to significantly improve the spatial resolution of EEG potential distributions by reducing head volume conductor effects and cancelling the influence of the electrical reference.

ERS/ERD was computed for correctly solved trials in the theta (4–7 Hz) and alpha (8–12 Hz) frequency band. We selected the broad alpha frequency range because an analysis with separate lower (8–10 Hz) and upper (10–12 Hz)

alpha bands (cf. Klimesch 1999) did not reveal task-related functional dissociations (similar to Bastiaansen et al. 2002; Stipacek et al. 2003). For ERS/ERD computation, the data from 500 to 2,500 ms after trial onset (during the fixation interval) served as reference interval. The activation interval comprised the entire time period from problem presentation (at 3,000 ms) until 125 ms before the onset of the oral response as registered by the voice-key. Consequently, the length of the activation interval varied between individuals and across trials and captured the entire period of arithmetic problem solving. The last 125 ms (i.e., 32 samples) of the response time were discarded to account for the delay of the voice-key trigger signal and to eliminate motor- and speech-related artefacts. The band power of reference (R) and activation (A) intervals were obtained by band-pass filtering, squaring the artefact-free EEG signal and averaging over all trials separately for R and A. The amount of ERS/ERD was calculated according to the formula: $\%\text{ERS/ERD} = [(A - R)/R] \times 100$. Positive values indicate increases in band power (ERS) and negative values indicate decreases (ERD). For statistical analyses, the %ERS/ERD values were topographically aggregated (by using the arithmetic mean) to obtain eight cortical areas per hemisphere (electrode positions given exemplarily for the left hemisphere): anteriofrontal (AF; Fp1, AF7, AF3), frontal (F; F7, F5, F3, F1), frontocentral (FC; FC5, FC3, FC1), central (C; C5, C3, C1), centroparietal (CP; CP5, CP3, CP1), parietal (P; P7, P5, P3, P1); parietooccipital (PO; PO7, PO3, O1), and temporal (T; FT7, T7, TP7).

In all statistical analyses, degrees of freedom were corrected for violations of the sphericity assumption by means of the Huynh–Feldt procedure; the probability of a type I error was maintained at 0.05. If applicable, uncorrected *df* values together with the corrected *P* value and the Huynh–Feldt epsilon (ε) are reported. Post hoc *t* tests were Bonferroni corrected.

Results

Behavioural data

The mean error rates and reaction times per problem type and operation are displayed in Fig. 1. A repeated measures ANOVA with problem type (retrieval, simple procedural, complex procedural) and operation (addition, subtraction) as within-subject factors was conducted on the error rates and reaction times. With regard to error rate, there were main effects of problem type [$F(2,38) = 76.94$, $P < 0.001$] and operation [$F(1,19) = 10.44$, $P < 0.01$], and a problem type × operation interaction $F(2,38) = 10.41$, $P < 0.01$. Post hoc *t* tests revealed that fewer errors were made on retrieval than on the procedural problems (Ps < 0.01 corrected). There were no differences between simple and complex procedural problems in addition, while in subtraction more errors were made on the complex than on simple procedural problems ($P < 0.001$ corrected). No differences in error rates between addition and subtraction for both retrieval and simple procedural problems were found, whereas for complex procedural problems more errors were made on additions than on subtractions ($P < 0.001$ corrected).

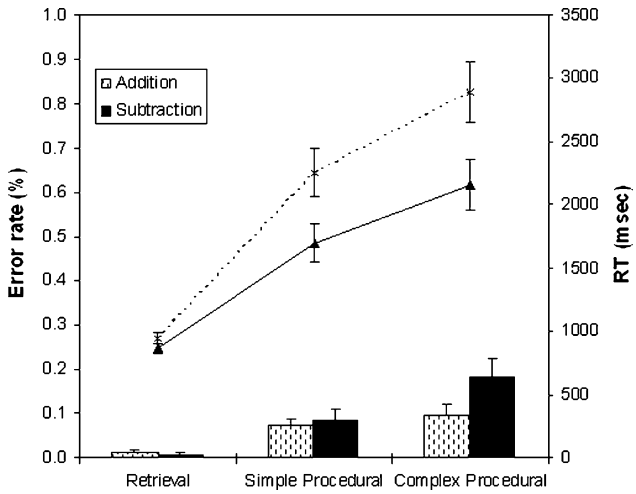

Fig. 1 Mean error rates and reaction times as a function of problem type and operation. *Bars* depict error rates (*left y*-axis) and lines represent reaction times (*right y*-axis). *Error bars* depict 1 SE of the mean

Turning to reaction time, there was a main effect of problem type [$F(2,38) = 212.67$, $P < 0.0001$], showing that retrieval problems were solved faster than simple procedural problems, which were solved faster than complex procedural problems. There was a main effect of operation [$F(1,19) = 104.95$, $P < 0.001$) and a problem type × operation interaction [$F(2,38) = 29.82$, $P < 0.0001$]. Post hoc *t* tests revealed that retrieval problems were solved equally fast in addition and subtraction, while procedural additions were solved faster than procedural subtractions (Ps < 0.001 corrected).

Event-related (de-) synchronisation (ERS/ERD)

Repeated measures ANOVAs were computed for the theta and alpha frequency band separately with problem type, operation, hemisphere (left vs. right), and topographical area (eight areas as described above) as within-subject factors.

In the theta band, an interaction of problem type × area × hemisphere [$F(14,266) = 2.28$, $P < 0.01$] was found. Differences between the three problem types mainly emerged in the left hemisphere with the retrieval problems showing the highest theta ERS (Fig. 2a). Post hoc comparisons revealed that the retrieval problems elicited significantly

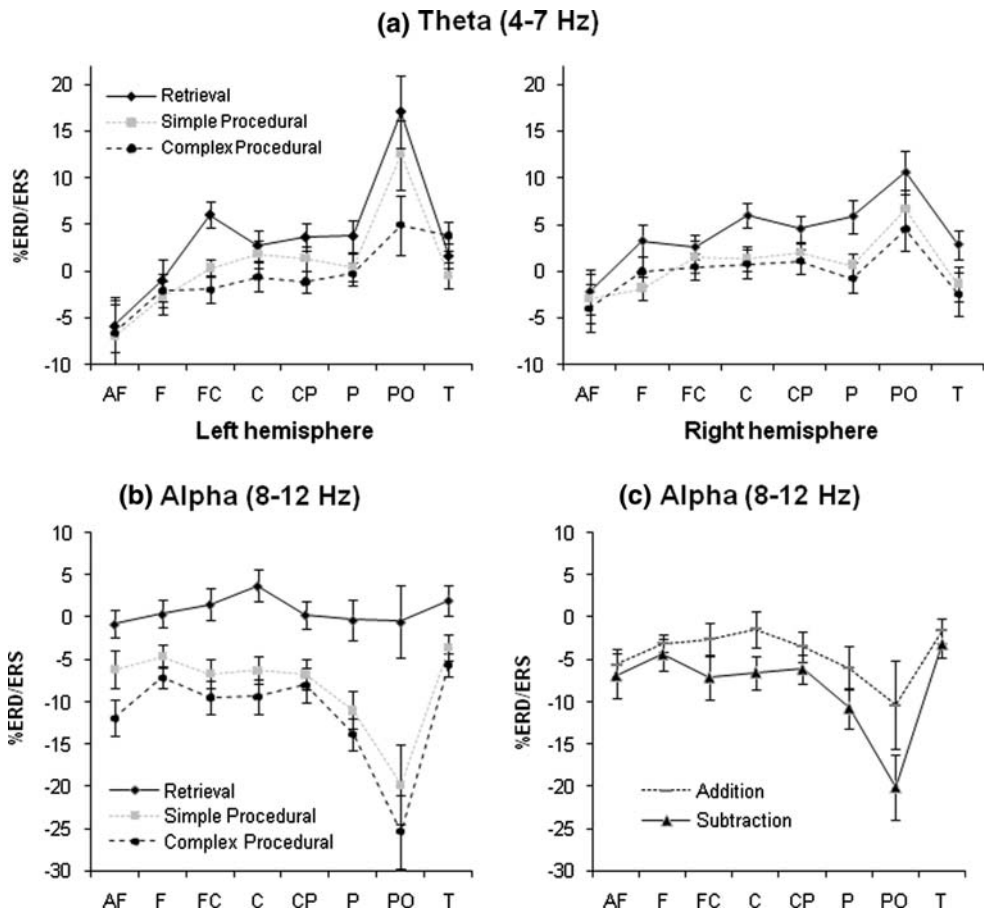

Fig. 2 Event-related (de-)synchronisation (%ERD/ERS) in the theta band (*upper panel*) and the alpha band (*lower panel*). *Error bars* depict 1 SE of the mean. *AF* anteriofrontal, *F* frontal, *FC* frontocentral, *C* central, *CP* centroparietal, *P* parietal, *PO* parietooccipital, *T* temporal. **a** Effects of problem type on theta ERS. Retrieval problems displayed significantly higher ERS in left FC and PO than complex procedural problems. **b** Effects of problem type on alpha ERD. Complex procedural problems displayed significantly stronger ERD than retrieval problems in all areas. **c** Effects of arithmetic operation on alpha ERD. The differences did not reach significance in the post hoc comparisons

higher theta ERS than the complex procedural problems in left frontocentral and parietooccipital areas (Ps < 0.05 corrected), and that the simple procedural problems showed stronger theta ERS than the complex procedural problems in the left parietooccipital cortex ($P < 0.05$ corrected). There were also main effects of problem type [$F(2,38) = 9.93$, $P < 0.001$, $\varepsilon = 0.81$] and area [$F(7,133) = 9.81$, $P < 0.001$, $\varepsilon = 0.44$], and an area × hemisphere interaction [$F(7,133) = 3.51$, $P < 0.01$, $\varepsilon = 0.61$]. Overall, the highest theta ERS was found for the retrieval problems (3.86%), followed by simple (0.75%) and complex procedural problems (−0.25%). Furthermore, theta ERS peaked in the parietooccipital area, most prominently in the left hemisphere. No operation effects emerged in the theta band.

In the alpha band, both problem- and operation-related effects were found. The problem type × area interaction [$F(14,266) = 9.09$, $P < 0.001$, $\varepsilon = 0.49$) revealed distinctive alpha ERD for simple and complex procedural problems, whereas the retrieval problems were accompanied by ERS/ERD values around 0 (Fig. 2b). More specifically, the ERD of retrieval problems was significantly lower than that of complex procedural problems in all topographical areas (Ps < 0.05 corrected). This difference was especially pronounced in the parietooccipital cortex bilaterally. The same pattern was found for main effects of problem type [$F(2,38) = 36.05$, $P < 0.001$), area [$F(7,133) = 9.48$, $P < 0.001$, $\varepsilon = 0.33$], and hemisphere [$F(1,19) = 6.67$, $P < 0.05$).

We also observed a significant operation × area interaction [$F(7,133) = 3.86$, $P < 0.05$, $\varepsilon = 0.45$]. Subtractions displayed higher alpha ERD than additions, with the largest difference in the parietooccipital cortex (Fig. 2c). These differences, however, failed to reach significance in the post hoc comparisons.

Discussion

The present high-resolution EEG study analysed the ERS and ERD in the theta and alpha frequency bands while solving different types of additions and subtractions to unravel the functional brain dynamics associated with the application of retrieval and procedural arithmetic strategies. To distinguish between these strategies, we presented problems with a high probability of retrieval or procedural strategy use. Retrieval problems had a sum or minuend below or equal to 10, which showed high frequencies of retrieval use in educated adults (e.g., Campbell and Xue 2001); procedural problems involved two-digit calculations,

to which the answer is unlikely to be directly retrieved from memory. Behavioural data revealed that retrieval problems were solved more accurate and faster than procedural problems, confirming the distinction between arithmetic fact recall and real calculation. The distinction between retrieval and procedural problems was also evident in the ERS/ERD patterns in the theta and the alpha frequency band.

As expected, the solution of retrieval problems elicited stronger theta ERS than procedural problems with significant differences in the frontocentral and parietooccipital brain areas of the left hemisphere, thereby extending previous EEG studies in arithmetic (Earle et al. 1996; Harmony et al. 1999; Micheloyannis et al. 2005). Considerable evidence indicates that oscillations in the theta frequency band play an important role in memory encoding and retrieval as they are thought to be involved in functionally linking the neocortex with the medial temporal lobe, especially the hippocampus (Bastiaansen and Hagoort 2003; Klimesch et al. 2005). Language studies have recently attributed theta ERS to the retrieval of lexical information from long-term memory (Bastiaansen et al. 2006). For example, Bastiaansen et al. (2005) found larger left temporal theta ERS while reading semantically rich (e.g., nouns) compared to semantically poor words (e.g., prepositions) which was interpreted to reflect the activation of a network involved in retrieving the lexical-semantic properties in the first class of words (see also Hald et al. 2006). Our findings of higher theta ERS in retrieval compared to procedural problems may similarly reflect the stronger reliance on retrieval of lexical information from long-term memory. In line with this, simple arithmetic problems are assumed to be stored verbally in long-term memory and to engage left-hemispheric language regions during retrieval (Dehaene et al. 2003; Grabner et al. 2009). The restriction of the theta effect to the left frontocentral and parietooccipital brain regions fits with this and suggests that the theta frequency band is sensitive not only to lexical-semantic retrieval during language processing but also to the retrieval of verbally represented arithmetic facts.

It should be noted that task-related increases in frontal theta ERS have also been associated with working memory load and attention demands (Gevins et al. 1997; Jensen and Tesche 2002; Kahana et al. 2001). Our findings, however, are not in line with this interpretation. If frontal theta ERS would reflect working memory processes, then this effect should be most prominent in the (complex) procedural problems, who have the highest working memory load. However, we found that theta ERS in left frontocentral and parietooccipital areas was higher in retrieval than procedural problems. Moreover, there was no effect of arithmetic operation in the theta band, even though the procedural subtraction problems were more difficult than the addition problems (Fig. 1), the latter of which suggests more effortful processing or working memory load in subtraction than in addition. This all indicates that the observed theta effects likely reflect the differential demands on lexical retrieval rather than on working memory resources.

Effects related to problem type were also found in the alpha frequency range. As hypothesised, procedural problems were accompanied by stronger alpha ERD than retrieval problems in all cortical areas, with the largest difference emerging in parietooccipital regions, bilaterally. Variations in parietal alpha ERD between different types of arithmetic problems have not been reported so far. Alpha oscillations are assumed to be modulated by thalamo-cortical and cortico-cortical loops (Klimesch et al. 2007) and have been observed to desynchronise while processing various cognitive tasks (Neubauer et al. 2006). It is assumed in current neurophysiological models that the alpha amplitude is inversely related to the activated cortical neuronal populations, and alpha ERD can be regarded as a correlate of cortical activation (Pfurtscheller and Lopes da Silva 1999, 2005). Several studies have corroborated this by showing higher alpha ERD over task-related areas in conditions with increasing difficulty or cognitive demands (e.g., Gevins et al. 1997; Stipacek et al. 2003). Thus, the present alpha ERD result may reflect higher cortical activation in the more demanding procedural compared to the simple retrieval problems. The largest activation difference over parietooccipital regions converges with previous fMRI studies showing stronger bilateral parietal activation in complex (vs. simple) arithmetic problems (Kong et al. 2005; Menon et al. 2000) and procedural (vs. retrieval) problems (Grabner et al. 2009).

The differential sensitivity of the theta and alpha band to arithmetic problem solving strategies is striking. Solving complex procedural problems is accompanied by clear-cut alpha ERD (most strongly at parietooccipital areas bilaterally) while no theta ERS was observed in any of the analysed brain areas. Conversely, the retrieval problems elicit strong theta ERS (at left frontocentral and parietooccipital areas) but no alpha ERD. This frequency-related distinction between retrieval and procedural strategies in arithmetic highlight the ERS/ERD method as an interesting tool to validate the often controversial use of verbal self-report measures to study arithmetic strategies (Kirk and Ashcraft 2001).

Our study also aimed to investigate neurophysiological differences between addition and subtraction problems. The triple-code model (Dehaene and Cohen 1997) assumes differences between simple addition, which should rely on a verbal representation, and simple subtraction, which should rely on a nonverbal quantity code. Our findings are not entirely in line with this assumption. There were no performance differences for the retrieval problems and no operation

effects in the theta band, suggesting similar underlying cognitive processes. Some evidence for stronger alpha ERD while solving subtraction compared to addition problems was found, but this difference failed to reach significance in the post hoc tests. Subtractions displayed higher ERD than additions at parietooccipital brain areas. This higher alpha ERD in subtraction may reflect the higher engagement of task-related cortical resources. These findings echo the fMRI data of Kong et al. (2005) who showed that both subtraction and procedural complexity activated left intraparietal sulcus and hypothesised that their common activation represents a general network that supports more difficult calculation.

To conclude, the present high-resolution EEG study shows a high sensitivity of the ERS/ERD measure to capture different strategies involved in solving addition and subtraction problems. The findings suggest that arithmetic fact retrieval is reflected in power increases (ERS) in the theta frequency band, whereas the application of more demanding procedural strategies is accompanied by power decreases (ERD) in the alpha band. The surprisingly strong frequency specificity of the strategy-related effects provides a potential starting point for the development of electrophysiological indices of different problem solving strategies in mental arithmetic.

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